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PRACTICAL 7: ROUTER CONFIGURATION AND INTER-NETWORK COMMUNICATION

Objective

Configure routers and establish communication between networks.

Equipment

- Cisco 1941 Router
- Cisco ISR4331 Router
- Cisco Switches
- PCs

Activities

Create two networks:

Network A

- 192.168.10.0/24

Network B

- 192.168.20.0/24

Configure router interfaces.

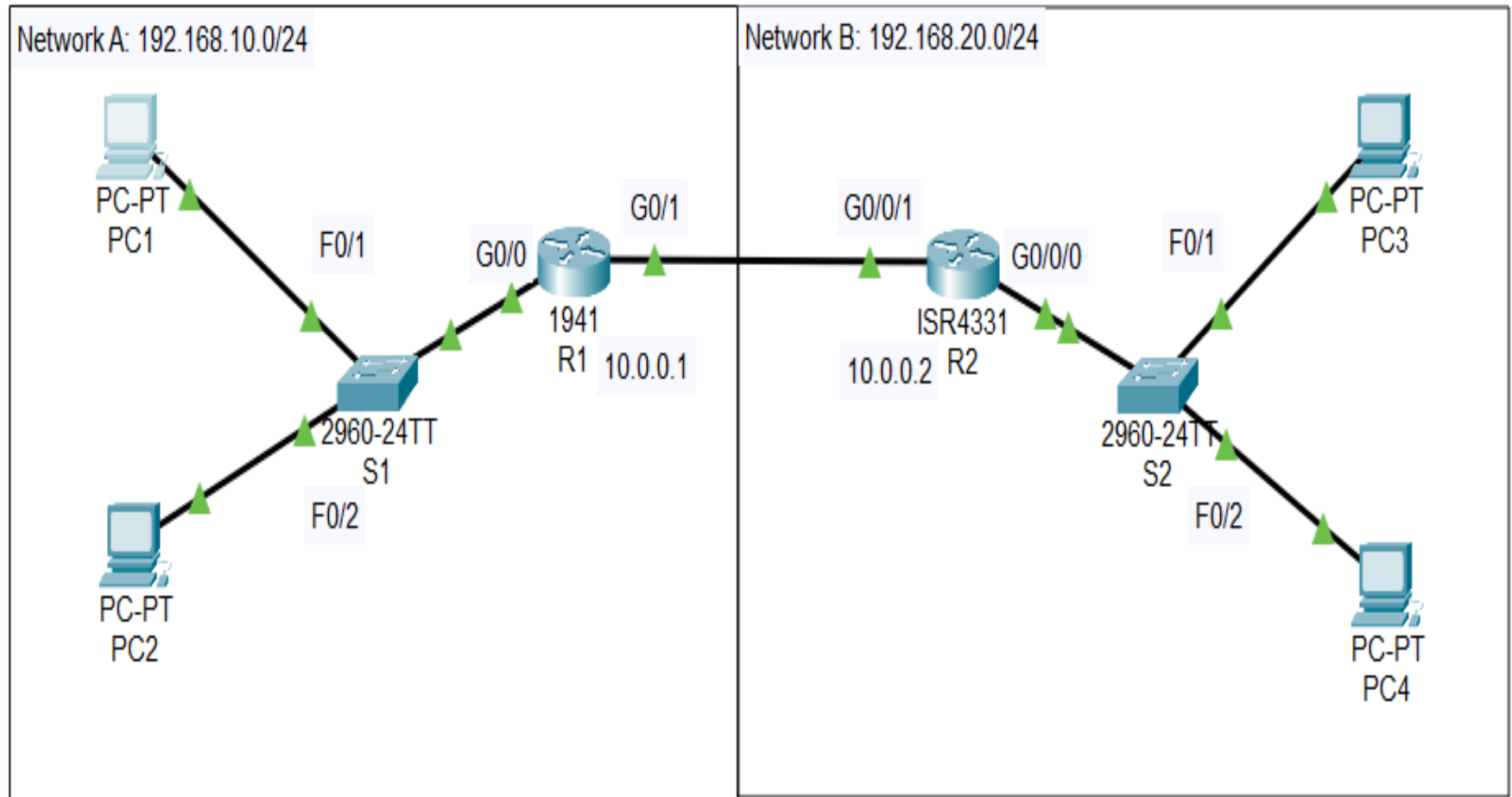
Example Configuration

- interface g0/0
- ip address 192.168.10.1 255.255.255.0
- no shutdown
- interface g0/1
- ip address 192.168.20.1 255.255.255.0
- no shutdown

Verification Commands

- show ip interface brief
- ping
- tracer

Network Topology:



Router 1 configuration:

```
Router>enable
Router#confi ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0

Router(config-if)#exit
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to

Router(config-if)#exit
Router(config)#
```

Static Route

```
Router(config-if)#exit
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#
```

Show ip interface router 1:

```
Router#show ip interface brief
Interface          IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0 192.168.10.1    YES manual up                    up
GigabitEthernet0/1 10.0.0.1        YES manual up                    up
Vlan1              unassigned      YES unset  administratively down down
Router#
```

Router 2 configuration:

```
Router>enable
Router#confi ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

Router(config-if)#exit
Router(config)#interface g0/0/1
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
```

Static route:

```
Router(config-if)#exit
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#
```

Show ip interface router 2:

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0     192.168.20.1    YES manual  up          up
GigabitEthernet0/0/1     10.0.0.2        YES manual  up          up
GigabitEthernet0/0/2     unassigned      YES unset   administratively down down
Vlan1                    unassigned      YES unset   administratively down down
```

PC addressing:

PC1

☐ DHCP	☒ Static
IPv4 Address	192.168.10.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

PC2

☐ DHCP	☒ Static
IPv4 Address	192.168.10.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0

PC3

☐ DHCP	☒ Static
IPv4 Address	192.168.20.10
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0

PC4

☐ DHCP	☒ Static
IPv4 Address	192.168.20.11
Subnet Mask	255.255.255.0
Default Gateway	192.168.20.1
DNS Server	0.0.0.0

Verification commands:

Router 1→ Router 2

```
Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
```

Router 2→ Router 1

```
Router#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

PC1→PC3

```
C:\>PING 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=13ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126
Reply from 192.168.20.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 4ms
```

PC4→PC2

```
Cisco Packet Tracer PC Command Line 1.0
C:\>PING 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.11: bytes=32 time<1ms TTL=126
Reply from 192.168.10.11: bytes=32 time<1ms TTL=126
Reply from 192.168.10.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Tracert PC1→PC3

```
C:\>TRACERT 192.168.20.10
```

```
Tracing route to 192.168.20.10 over a maximum of 30 hops:
```

1	0 ms	0 ms	0 ms	192.168.10.1
2	0 ms	0 ms	0 ms	10.0.0.2
3	0 ms	0 ms	0 ms	192.168.20.10

Deliverable

Successful communication between both networks.

Assessment Rubric (10 Marks)

Criteria	Marks
Router configuration	4
Interface addressing	2
Connectivity testing	2
Documentation	2
Total	10